BANK CUSTOMER SEGMENTATION AND TRANSACTION ANALYSIS

Business Intelligence & Exploratory Data Analysis Report

Interactive Power BI dashboard for customer demographics, transaction behavior, segmentation, and profitability/risk indicators

Project Snapshot	Value
Data source	bank_transactions.csv
Dashboard pages	4
Primary tool	Microsoft Power BI Desktop
Analysis type	Exploratory and descriptive analysis
Project author	Afrin

Prepared as a portfolio-ready project document

1. Executive Summary

This project presents an interactive Power BI dashboard for exploring bank customer demographics, transaction behavior, customer segments, and profitability-related and risk indicators. The report is organized into four pages: Customer Demographics Analysis, Transaction Behavior Insights, Customer Segmentation Analysis, and Profitability & Risk Indicators.

The dashboard is intended to make data exploration more accessible through visual summaries and interactive filtering. Exact numeric findings should be taken from the final PBIX report after validating the data, measures, and active filters.

Project metric	Status / description
Source dataset	bank_transactions.csv
Dashboard sections	4
Report format	Power BI (.pbix)
Purpose	Descriptive analysis and exploratory business insights

2. Business Problem

Customer and transaction data can be difficult to interpret quickly when stored only in raw tabular form. A consolidated dashboard can help users explore customer distributions, compare transaction patterns, and review differences between customer groups.

Business questions

• What demographic patterns are visible in the customer base?

• How does transaction activity vary across available time periods and categories?

• How do customer segments compare on available measures?

• What profitability-related measures and risk indicators can be explored from the data?

3. Objectives

• Analyze available customer demographic attributes.

• Explore transaction behavior and time-based patterns.

• Compare customer segments using the available fields and definitions.

• Present profitability-related measures and risk indicators where supported by the dataset.

• Build a clear, interactive, multi-page Power BI report.

• Communicate findings while distinguishing observed patterns from causal claims.

4. Data Source & Methodology

4.1 Data source

The primary input file is data/bank_transactions.csv. The dataset's exact row count, date range, schema, and original source should be documented from the file and source documentation; these details are not assumed in this report.

4.2 Analytical workflow

Stage	Description
Data import	Load the CSV into Power BI Desktop.
Data preparation	Inspect field types, missing values, duplicates, and inconsistent categories.
Data modelling	Validate relationships and fields used by the report, if applicable.
Measures	Create or validate calculations and KPI definitions.
Visualization	Build the four dashboard pages around the project objectives.
Validation	Check totals, filters, labels, and measures against the source data.

4.3 Data pipeline

Pipeline
CSV data → Data inspection → Cleaning and preparation → Measures / model → Dashboard design → Validation and reporting

5. Dataset Profile

The project uses a CSV dataset containing banking transaction records. Before final submission, record the actual number of rows, columns, date coverage, and key field names from the source file.

Dimension	Details to confirm
Records	Read from the CSV
Fields	List the actual column names
Date coverage	Minimum and maximum date, if a date field exists
Missing values	Profile important fields
Duplicates	Check transaction identifiers or repeated records
Data types	Confirm dates, numeric measures, and categories

6. Data Cleaning & Analytical Preparation

• Assign suitable data types to numeric, date, and categorical fields.

• Inspect missing values and decide how to handle them based on business meaning.

• Check duplicate records before removing anything.

• Standardize category labels where inconsistent values occur.

• Validate unusual values and ensure measures use the intended units.

• Document transformations so the report can be refreshed consistently.

6.1 Quality checks

Data preparation should preserve legitimate records and avoid silently removing outliers or unusual transactions. Any exclusions or replacements should be documented and justified.

7. Dashboard Analysis

7.1 Customer Demographics Analysis

This page explores customer characteristics such as age, gender, and location where these fields are available. Category and age-group comparisons can help describe the composition of the customer base. Insert the final dashboard screenshot and validated demographic counts here.

7.2 Transaction Behavior Insights

This page examines available transaction measures and time-based patterns. Charts may be used to compare transaction volume or amount across relevant dates, times, and categories. Any peak period should be reported only after confirming it against the final report filters.

7.3 Customer Segmentation Analysis

This page compares customer groups based on the segmentation logic implemented in the PBIX file. Segment labels, including labels such as loyal, new, or lost, should be described only if they are present in the model and supported by documented rules.

7.4 Profitability & Risk Indicators

This page presents the profitability-related measures and risk indicators included in the dashboard. Any classification, threshold, or financial interpretation should be based on documented definitions and validated source fields. A dashboard indicator alone does not establish creditworthiness or predict default.

8. Interactivity and User Experience

The report is structured as a multi-page interactive experience. Users can navigate between the four analytical views and use the slicers, filters, and visual interactions that are configured in the PBIX file. These features support focused exploration without manually filtering the source CSV.

9. Key Observations

The four-page design supports analysis from complementary perspectives: customer composition, transaction patterns, differences among customer segments, and profitability/risk-related measures. Specific values, percentages, rankings, or trends are intentionally not invented in this report. Add validated observations from the final dashboard, including the date range and filters used.

10. Business Insights

• Demographic analysis can help describe who is represented in the customer base.

• Transaction analysis can reveal patterns that merit further investigation.

• Segment comparisons can support discussions about engagement and customer value when segment definitions are clear.

• Profitability and risk indicators require documented formulas and appropriate interpretation.

• Visual patterns are descriptive; they do not independently prove causation or predict future outcomes.

11. Limitations & Scope

• Results depend on source-data completeness and accuracy.

• Exact dataset statistics and KPI definitions must be verified from the final project files.

• Customer segments are only meaningful when their assignment rules are documented.

• Risk analysis is limited to the available fields and measures.

• Historical or descriptive patterns should not be treated as validated forecasts or as the sole basis for financial decisions.

12. Recommendations & Future Enhancements

• Add a data dictionary and definitions for each KPI.

• Document the source, row count, columns, and refresh date.

• Add screenshots of each dashboard page to the repository.

• Validate segment rules and key metrics against the underlying data.

• Include additional drill-through views or time comparisons where useful.

• Add concise business recommendations supported by measured results.

13. Reproducibility & Project Assets

Asset	Purpose
data/bank_transactions.csv	Input transaction dataset
powerbi/Bank Customer Segmentation.pbix	Interactive Power BI report
screenshots/	Optional dashboard screenshots
README.md	Project overview and instructions
.gitignore	Excludes selected temporary or local files

14. How to Open the Project

• Download or clone the repository from GitHub.

• Open powerbi/Bank Customer Segmentation.pbix in Microsoft Power BI Desktop.

• If prompted, update the source path to data/bank_transactions.csv.

• Refresh the data if required and review all four dashboard pages.

15. Conclusion

The Bank Customer Segmentation & Transaction Analysis project demonstrates how Power BI can organize banking transaction data into an interactive analytical report. The dashboard brings together customer demographics, transaction behavior, segmentation, and profitability/risk-related indicators. Validating the dataset, metric definitions, and final numerical findings will strengthen the reliability and usefulness of the report.

Project Repository

https://github.com/afrin277/Bank_Customer_Segmentation

Appendix A — Final Submission Checklist

• Confirm the dataset source and usage terms.

• Record the actual row count, column names, and date range.

• Insert screenshots of all four dashboard pages.

• Verify numerical observations and KPI definitions against the PBIX file.

• Check that the GitHub repository contains the CSV, PBIX, README, and any intended supporting assets.